Recovering Muscle Weakness from Kinematics Alone in a Musculoskeletal Arm Simulation: State/Action Spaces, Reference Generation, and Identification

myoArm identifiability study — technical report

2026-06-11

Abstract

We study *sim-internal identifiability*: in a 63-muscle MuJoCo arm (myoArm), we scale per-muscle maximum isometric force (s_F) and optimal fiber length (s_L), let a reinforcement-learning policy *softly* track human elevation references, and then recover (s_F, s_L) from the resulting motion. We give formal definitions of the (i) Markov decision process state, observation, and action spaces; (ii) the reward; (iii) the policy/value output spaces and PPO objective; (iv) two generative models for the reference trajectories (functional PCA / ProMP and a VAE) with their input/output spaces; and (v) the time-series identification regressor. Using **kinematics only** (no EMG), a temporal convolutional regressor recovers shoulder/elbow s_F at $R^2 = 0.81$; unlocking the wrist with the full movement adds wrist-muscle identifiability ($R^2 \approx 0.8$) at the cost of shoulder/elbow accuracy.

1 Problem formulation

Let the plant be the myoArm model with generalized coordinates $\mathbf{q} \in \mathbb{R}^{n_q}$ (n_q=38: 27 independent + 11 scapular-rhythm coordinates coupled by equality constraints), generalized velocities $\dot{\mathbf{q}} \in \mathbb{R}^{n_v}$, and n_μ=63 Hill-type muscle actuators with activation states $\mathbf{a}^{\text{act}} \in [0,1]^{n_\mu}$. A *perturbation* (muscle weakness) is a pair of channel-wise scale vectors

$$\boldsymbol{\theta} = (\mathbf{s}_F, \mathbf{s}_L) \in \Theta := [0.05, 1]^{|\mathcal{C}|} \times [0.70, 1.20]^{|\mathcal{C}|}, \tag{1}$$

applied to $|\mathcal{C}|$ *perturbation channels* (groups of muscles). For channel c acting on muscle i, the MuJoCo gain parameters are scaled as

$$\text{gain}_{i,2} \leftarrow s_F^c \, \text{gain}^0_{i,2} \quad (\text{gain}_2 = F_{\max}), \qquad \text{gain}_{i,0:2} \leftarrow \text{gain}^0_{i,0:2}/s_L^c \quad (\text{operating length range}). \tag{2}$$

The goal is to learn an estimator $\widehat{\boldsymbol{\theta}} = g(\text{observed motion})$ and report per-channel recovery quality R^2_c. The motion is produced by a policy that softly tracks a reference trajectory $q^{\text{ref}} : [0, H] \to \mathbb{R}^{|\mathcal{D}|}$ over the tracked degrees of freedom $\mathcal{D}$.

2 Markov decision process

The control MDP is $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathcal{A}, P, r, \gamma)$ at control rate 50 Hz (Δt=0.02 s, physics substeps F=10 at 500 Hz), horizon H (175 for elevation-rise, 200 for the full sequence).

2.1 State space $\mathcal{S}$

The (Markov) state augments the physical state with the episode-fixed context:

$$\mathcal{S} = \underbrace{\mathbb{R}^{n_q}}_{\mathbf{q}} \times \underbrace{\mathbb{R}^{n_v}}_{\dot{\mathbf{q}}} \times \underbrace{[0,1]^{n_\mu}}_{\mathbf{a}^{\text{act}}} \times \underbrace{\Theta}_{\text{weakness}} \times \underbrace{\Lambda}_{\text{latent}} \times \underbrace{\{T_1, T_2\}}_{\text{task}} \times \underbrace{\{0, \dots, H\}}_{\text{step } t}, \tag{3}$$

where $\Lambda \subset \mathbb{R}^5$ collects the movement-variation latent $\boldsymbol{\ell} = (T, \text{peak}, \text{skew}, \text{plane}, \text{elbow})$ (Sec. 4). $\boldsymbol{\theta}, \boldsymbol{\ell}$, task are drawn at reset and held fixed within an episode; q^{ref} is a deterministic function of $(\text{task}, \boldsymbol{\ell})$.

2.2 Observation space $\mathcal{O}$

The policy does not see $\mathcal{S}$ directly but an observation $\mathbf{o}_t = \phi(s_t) \in \mathcal{O} \subset \mathbb{R}^{d_o}$, $d_o = 76$ (wrist-unlocked mode: 96). With tracked DoF $\mathcal{D} = \{\text{elv}, \text{plane}, \text{elbow}, \text{prosup}\}$ ($|\mathcal{D}|$=4) and free DoF rot, and writing $\mathbf{a} \in \mathcal{A}$ for the previous action, ϕ concatenates

$$\begin{aligned}\mathbf{o}_t = \Big[\ & \tfrac{1}{\pi}\, \mathbf{q}_{\mathcal{D}\cup\{\text{rot}\}}\ (5), \quad \tfrac{1}{5}\, \dot{\mathbf{q}}_{\mathcal{D}\cup\{\text{rot}\}}\ (5), \quad \mathbf{a}^{\text{act}}_{\mathcal{A}}\ (26), \\ & \tfrac{1}{\pi} q^{\text{ref}}_{\mathcal{D}}(t)\ (4), \quad \tfrac{1}{\pi}\big(q^{\text{ref}}_{\mathcal{D}}(t) - \mathbf{q}_{\mathcal{D}}\big)\ (4), \quad \tfrac{1}{\pi} q^{\text{ref}}_{\mathcal{D}}(t{+}5)\ (4), \\ & \big[\varphi_t,\ \mathbb{1}_{\text{hold}}\big]\ (2), \quad \tilde{\boldsymbol{\ell}}\ (5), \quad [\mathbf{s}_F, \mathbf{s}_L]\ (2|\mathcal{C}|{=}20), \quad \mathbb{1}[\text{task}{=}T_2]\ (1)\ \Big],\end{aligned} \tag{4}$$

with phase $\varphi_t = t/H$, $\mathbb{1}_{\text{hold}} = \mathbb{1}[t \geq T/\Delta t]$, and normalized latent $\tilde{\boldsymbol{\ell}} = (T/3, \text{peak}/120, \text{skew}/2, \text{plane}/45, \text{elbow}/130)$. Dimension check: 5+5+26+4+4+4+2+5+20+1 = 76. A running-statistics affine normalizer ν (VecNormalize, clipped to $[-10, 10]$) is applied before the network: $\tilde{\mathbf{o}}_t = \text{clip}\big((\mathbf{o}_t - \boldsymbol{\mu}_o)/\boldsymbol{\sigma}_o,\ -10, 10\big)$. The block $[\mathbf{s}_F, \mathbf{s}_L]$ encodes the (privileged) weakness, justified by the *adapted chronic* scenario: the CNS has re-adapted to fixed muscle parameters, so motor commands reflect $\boldsymbol{\theta}$.

2.3 Action space $\mathcal{A}$ and dynamics

$$\mathcal{A} = [-1, 1]^{|\mathcal{A}|}, \qquad |\mathcal{A}| = 26 \text{ (shoulder 15 + elbow 9 + pronators PT,PQ)}, \tag{5}$$

mapped to muscle excitation $u \in [0,1]^{|\mathcal{A}|}$ with signal-dependent motor noise (Harris–Wolpert):

$$u_j = \tfrac{1}{2}(a_j + 1), \qquad u_j \leftarrow \text{clip}\big(u_j + \eta_j, 0, 1\big), \quad \eta_j \sim \mathcal{N}\big(0, (\sigma_{\text{sd}} u_j + \sigma_0)^2\big), \tag{6}$$

σ_{sd}=0.1, σ_0=0.01. Non-action muscles take u=0. MuJoCo maps excitation to activation through first-order dynamics (τ_{act}=10 ms, τ_{deact}=40 ms) and force through a Hill force–length–velocity model; transition P integrates F physics substeps.

2.4 Reward

$$r(s_t, \mathbf{a}_t) = \text{TASK}_t \cdot \text{QUALITY}_t + w_e\, \text{EFFORT}_t + \text{pen}^{\text{rot}}_t, \tag{7}$$

$$\text{TASK}_t = \sum_{i \in \mathcal{D}} w_i\ \exp\Big(- k^{\text{out}}_i \big[\max(0,\ |q_i - q^{\text{ref}}_i| - b_i)\big]^2 \Big), \tag{8}$$

$$\text{QUALITY}_t = \exp\Big(- \big(c_J\, \overline{\dddot{q}^2} + c_D\, \overline{(\mathbf{a}_t - \mathbf{a}_{t-1})^2} + c_S\, \mathbb{1}_{\text{hold}}\, \overline{\dot{q}^2}\big) \Big), \tag{9}$$

$$\text{EFFORT}_t = -\tfrac{1}{|\mathcal{A}|} \textstyle\sum_j (a^{\text{act}}_j)^p, \qquad \text{pen}^{\text{rot}}_t = -w_r \big[\max(0, |q_{\text{rot}}| - 50^\circ)\big]^2, \tag{10}$$

where $\overline{(\cdot)}$ is the mean over $\mathcal{D}$. The two-sided *soft band*: deviations within b_i incur *no* penalty (allowing self-paced variation), beyond b_i a Gaussian falloff; this preserves weakness as deviation rather than absorbing it. Coefficients (locked): $w = (0.15, 0.15, 0.25, 0.20)$, $b = (10, 15, 12, 20)^\circ$, $k^{\text{out}} = (30, 18, 22, 12)$, w_e=0.05, p=2, c_J=5×10^{-6}, c_D=0.5, c_S=1.0, w_r=0.5. Episode terminates (with $r-1$) if $|q_{\text{elv}} - q^{\text{ref}}_{\text{elv}}| > 50^\circ$, $|q_{\text{elbow}} - q^{\text{ref}}_{\text{elbow}}| > 60^\circ$, or $\max|\dot{q}| > 40$.

3 Policy, value, and PPO

3.1 Output spaces

The stochastic policy maps observations to a distribution over actions,

$$\pi_\psi : \mathcal{O} \to \mathcal{P}(\mathcal{A}), \qquad \pi_\psi(\cdot \mid \mathbf{o}) = \mathcal{N}\big(\boldsymbol{\mu}_\psi(\mathbf{o}),\, \mathrm{diag}(e^{2\boldsymbol{\sigma}_\psi})\big), \tag{11}$$

so the *policy output space* is $\mathbb{R}^{|\mathcal{A}|} \times \mathbb{R}^{|\mathcal{A}|}_{>0}$ (mean $\boldsymbol{\mu}_\psi(\mathbf{o}) \in \mathbb{R}^{26}$ and a state-independent log-std $\boldsymbol{\sigma}_\psi \in \mathbb{R}^{26}$). The value head outputs $V_\psi : \mathcal{O} \to \mathbb{R}$. Both $\boldsymbol{\mu}_\psi$ and V_ψ are MLPs with hidden layers $[256, 256]$ and tanh activations (separate networks).

3.2 Objective

With generalized advantage $\hat{A}_t = \sum_{l\geq 0} (\gamma\lambda)^l \delta_{t+l}$, $\delta_t = r_t + \gamma V_\psi(\mathbf{o}_{t+1}) - V_\psi(\mathbf{o}_t)$, and ratio $\rho_t(\psi) = \pi_\psi(\mathbf{a}_t \mid \mathbf{o}_t)/\pi_{\psi_{\text{old}}}(\mathbf{a}_t \mid \mathbf{o}_t)$, PPO minimizes

$$\mathcal{L}(\psi) = -\,\mathbb{E}_t\Big[\min\big(\rho_t \hat{A}_t,\ \mathrm{clip}(\rho_t, 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\Big] + c_v\,\mathbb{E}_t\big[(V_\psi(\mathbf{o}_t) - \hat{R}_t)^2\big] - c_e\,\mathbb{E}_t\big[\mathcal{H}(\pi_\psi(\cdot \mid \mathbf{o}_t))\big]. \tag{12}$$

Hyperparameters: ϵ=0.2, learning rate 3×10^{-4}, rollout $n_{\text{steps}}\times n_{\text{env}} = 512\times20$, minibatch 4096, 10 epochs/update, γ=0.99, λ=0.95, c_v=0.5, c_e=0, gradient-norm clip 0.5. Curriculum/warm-start: M1 healthy fixed (T1) $\to$ M2 latent distribution + T2 $\to$ M3 perturbation k=$1 \to k \leq 3$; weights are warm-started across stages ($\mathcal{O}$ dimension held fixed).

4 Reference (imitation target) generation

The reference q^{ref} is generated from human data (KIMHu, 20 subjects, Kinect-V2 3D joints). Per-repetition multichannel rise (or full-cycle) curves are derived *from 3D positions only* and resampled to N phase points, yielding a dataset

$$\mathcal{X} = \{\mathbf{X}^{(m)} \in \mathbb{R}^{N\times d}\}_{m=1}^{M}, \quad (N, d) = (50, 4) \text{ [rise] or } (80, 6) \text{ [full+wrist]},\ M \approx 200, \tag{13}$$

with channels (elevation, elbow, plane, pronation [+ wrist flexion, deviation]). After per-channel standardization, $\mathbf{x}^{(m)} = \mathrm{vec}\big((\mathbf{X}^{(m)} - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma}\big) \in \mathbb{R}^{Nd}$.

4.1 Functional PCA / ProMP (adopted)

Input space $\mathbb{R}^{Nd}$; **output** a sampled trajectory in $\mathbb{R}^{N\times d}$ (then time-warped to $\mathbb{R}^{H\times|\mathcal{D}|}$). Center $\mathbf{x}_c^{(m)} = \mathbf{x}^{(m)} - \bar{\mathbf{x}}$, take the SVD $\mathbf{X}_c = \mathbf{U}\mathbf{S}\mathbf{V}^\top$, eigenvalues $\lambda_k = S_{kk}^2/(M-1)$, principal functions $\boldsymbol{\phi}_k = \mathbf{V}_{:,k}$, and retain

$$K = \min\Big\{K :\ \textstyle\sum_{k\leq K}\lambda_k \big/ \sum_k \lambda_k \geq 0.95\Big\} \quad (K = 13\text{–}24 \text{ empirically}). \tag{14}$$

Standardized scores $\hat{w}_k^{(m)} = \langle \mathbf{x}_c^{(m)}, \boldsymbol{\phi}_k \rangle / \sqrt{\lambda_k}$. **Generation** (kernel density in score space; bootstraps a real score vector with Gaussian jitter, preserving non-Gaussian structure and the empirical covariance):

$$\hat{\mathbf{z}} = \hat{\mathbf{w}}^{(j)} + \boldsymbol{\varepsilon}, \quad j \sim \mathrm{Unif}\{1{:}M\},\ \boldsymbol{\varepsilon} \sim \mathcal{N}(0, h^2 I),\ h{=}0.35; \quad \hat{\mathbf{x}} = \bar{\mathbf{x}} + \sum_{k=1}^{K} \hat{z}_k \sqrt{\lambda_k}\, \boldsymbol{\phi}_k. \tag{15}$$

With $\hat{\mathbf{z}} \sim \mathcal{N}(0, I)$ the generated covariance equals the (truncated) empirical covariance $\sum_k \lambda_k \boldsymbol{\phi}_k \boldsymbol{\phi}_k^\top$, so dispersion is preserved (no mode shrinkage). A duration $T \sim \widehat{\mathrm{Emp}}\{T^{(m)}\}$ time-warps $\hat{\mathbf{X}}$ via $\varphi(t) = \min(t/T, 1)$ to give q^{ref}. Empirically peak 94±9° (real) vs. 95±9° (fPCA); marginal Kolmogorov–Smirnov distance 0.07–0.14.

4.2 Variational autoencoder (baseline)

Encoder $q_\vartheta : \mathbb{R}^{Nd} \to \mathcal{P}(\mathbb{R}^6)$ and **decoder** $p_\vartheta : \mathbb{R}^6 \to \mathbb{R}^{Nd}$ (latent space $\mathbb{R}^6$):

$$\boldsymbol{\mu}_z, \log \boldsymbol{\sigma}_z^2 = \mathrm{MLP}_{200\to128\to128\to(6,6)}(\mathbf{x}), \quad \hat{\mathbf{x}} = \mathrm{MLP}_{6\to128\to128\to200}(\mathbf{z}), \quad \mathbf{z} = \boldsymbol{\mu}_z + \boldsymbol{\sigma}_z \odot \boldsymbol{\varepsilon}, \tag{16}$$

trained by the β-ELBO (β=0.5)

$$\mathcal{L}_{\mathrm{VAE}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \beta \cdot \tfrac{1}{2} \textstyle\sum_k \left(\sigma_{z,k}^2 + \mu_{z,k}^2 - 1 - \log \sigma_{z,k}^2\right), \tag{17}$$

Adam (10^{-3}, 1500 epochs). Generation: $\mathbf{z} \sim \mathcal{N}(0, I) \to p_\vartheta$. The KL term plus small M induce *under-dispersion* (peak s.d. 9° → 4°) and partial posterior collapse (3/6 active dims); fPCA is therefore preferred.

5 Identification regressor

5.1 Input/output spaces

From $N \approx 16$–20k policy rollouts (motor noise on; deterministic policy mean), each episode yields a *kinematics-only* multivariate time series and a label:

$$g : \underbrace{\mathbb{R}^{T\times C}}_{\text{kinematics } \mathbf{Z}} \times \underbrace{\mathbb{R}^{6}}_{\text{known covariates } \boldsymbol{\ell}} \longrightarrow \underbrace{\mathbb{R}^{2|\mathcal{C}|}}_{\widehat{\boldsymbol{\theta}} = [\hat{\mathbf{s}}_F; \hat{\mathbf{s}}_L]}, \tag{18}$$

with $C = 2(|\mathcal{D}|+1) + |\mathcal{D}|$ channels (positions and velocities of tracked DoF + rotation, plus tracking errors): C=14 (base) or 20 (wrist). *No muscle activation / EMG is used.* Variable lengths are zero-padded to $T_{\max}$ with mask $\mathbf{m} \in \{0,1\}^T$.

5.2 Architecture (dilated TCN with masked pooling)

$$\mathbf{H}^{(0)} = \mathrm{ReLU}\,\mathrm{GN}\,\mathrm{Conv1d}_{k=7,s=2}(\mathbf{Z}^\top), \qquad \mathbf{H}^{(l)} = \mathrm{ReLU}\,\mathrm{GN}\,\mathrm{Conv1d}_{k=5,\,\mathrm{dil}=2^l}(\mathbf{H}^{(l-1)}),\ l{=}1,2,3, \tag{19}$$

$$\mathbf{g} = \left[\ \frac{\sum_t m_t \mathbf{H}_t}{\sum_t m_t}\ ;\ \max_{t: m_t = 1} \mathbf{H}_t\ ;\ \boldsymbol{\ell}\ \right], \qquad \widehat{\boldsymbol{\theta}} = \mathrm{MLP}_{\cdot\to128\to128\to2|\mathcal{C}|}(\mathbf{g}), \tag{20}$$

channel widths $64 \to 64 \to 128 \to 128$, GroupNorm, masked average+maximum temporal pooling. Loss with severity reweighting $w_n = 1 + 2\max_c(1 - s_F^{c,n})$ and Huber:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{n \in \mathcal{B}} w_n \cdot \frac{1}{2|\mathcal{C}|} \sum_o \mathrm{Huber}\big(\hat{\theta}_o^{(n)} - \theta_o^{(n)}\big), \tag{21}$$

Adam (10^{-3}, weight decay 10^{-5}) with cosine schedule, batch 256, 100–120 epochs, 85/15 hold-out (best-val). Evaluation: per-channel R^2 and MAE.

6 Results

Kinematics-only recovery of s_F (Fmax scale), fPCA references with motor noise (Table 1). The temporal model is essential: a snapshot (mean/max summary) MLP attains only $R^2 \approx 0.36$ on the same kinematics, versus 0.81 for the TCN — the weakness signal lives in the *temporal structure* (rise slope, deviation onset).

Setup	input ch. C	shoulder/elbow $R^2(s_F)$	wrist $R^2(s_F)$
rise, T1+T2, wrist locked	14	**0.81**	–
full-sequence + wrist, T1	20	0.60	0.81
full-sequence + wrist, T1+T2	20	0.55	0.74

Table 1: Per-configuration kinematics-only identifiability. Unlocking the wrist with the full movement adds wrist-muscle identification but dilutes shoulder/elbow accuracy, because the elevation-rise phase (muscles maximally loaded against gravity) is the most informative segment.

Per-channel (rise, best setup): A_lowcuff .94, B_latadd .88, SUPSP/BIClong/TRIlong .86–.87, DELT1 .84, DELT2 .79, DELT3 .75, CORB/PECM1 .62–.65. The length scale s_L is harder from kinematics alone ($R^2 \approx 0.56$).

Conclusions. (i) Muscle *force* weakness of shoulder/elbow muscles is strongly recoverable from movement kinematics alone (R^2=0.81), without EMG. (ii) Reference generation by fPCA/ProMP reproduces the empirical trajectory distribution faithfully (no under-dispersion), outperforming a VAE. (iii) The soft-band reward is the mechanism that turns weakness into an identifiable signal (mild $\to$ activation compensation, severe $\to$ kinematic deviation). (iv) There is a protocol trade-off: one movement cannot simultaneously maximize shoulder/elbow and wrist identifiability.